Skytree Server Installation

Release 15.4.0

SKYTREE®

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- Log in to support.skytree.net.
- Send questions to support@skytree.net.
- Call 408.392.9300 and select option 3.
- You can also view additional Skytree documentation by logging on to support.skytree.net.

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SKYTREE

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Chapter 1 Introduction

About this Document

Skytree Server can be installed on a single machine, in a distributed environment, or in a Hadoop/YARN environment.

Standalone (single machine) installation is described in *Standalone Installation* (page 7). Additional instructions are given in *Distributed Installation* (page 10) for distributed installations, and in *Installation for YARN/Hadoop* (page 13) for installations on Hadoop clusters.

Chapter 2 Requirements

Requirements are broken up into supported hardware and platforms. Additional dependencies also exist.

Refer to the following topics:

- Compute Hardware Specifications (page 3)
- Supported Platforms (page 4)
- Dependencies (page 4)

Compute Hardware Specifications

Skytree Server takes full advantage of compute, memory, and interconnect resources. We recommend the use of high-performance compute hardware to perform large machine learning tasks. We also recommend the use of dedicated machines for highest performance and low probability of outage or interruptions.

Compute Cores, Memory and Storage

We recommend at least 8-core Xeon 56xx series processors with at least 8 GB of (DDR3 or faster) memory per core. Dual-or quad-socket servers are also recommended, since high core counts per shared memory node are beneficial for many machine learning tasks. Any reasonable form of local storage will suffice for installation of Skytree Server software. Distributed Skytree Server jobs require a file system that is shared between compute nodes. The performance of even enterprise-grade network-attached-storage servers can be the bottleneck in multi-server/multi-user environments, and high-performance storage area networks should be considered. When running on Hadoop, the Hadoop Distributed File System (HDFS) provides a distributed file system service with a customizable replication factor for performance and redundancy. For non-Hadoop environments, HPC file systems (such as Lustre) with fast interconnects are recommended for best performance.

Interconnects and Topology

Most distributed machine learning tasks require some amount of inter-node communication (file I/O, load balancing or workload distribution), and the network interconnect can quickly become the bottleneck, especially when large messages (or files) have to transferred in a one-to-all or even all-to-all pattern. While Skytree Server will run on any interconnect speeds (as low as 100Mbit or 1Gigabit), we strongly recommend at least 10Gigabit network interconnects. This includes network interface cards (NICs) as well as supporting network switches and/or routers that support a high cross-sectional bandwidth and low latencies. Fat-tree network topologies are recommended for small to medium-size clusters, and hyper-torus or alternate cost-efficient topologies are most widely used for larger clusters.

Supported Platforms

- 64-bit (x86_64) Red Hat Enterprise Linux 6.x
- CentOS 6.x

> **Note:** If you intend to use Hadoop along with Skytree Server, consult the technical specifications of your Hadoop Distribution vendor for the operating system requirements of the specific version you intend to install.

Dependencies

For the Skytree Server data preparation and Hadoop tools, either the OpenJDK (http://openjdk.java.net) Development Kit 7 or Oracle Java (http://www.oracle.com/technetwork/java/index.html) SE 7 must be available. Please install either package as recommended for your Linux distribution.

Skytree Server also depends on OpenSSL Please install the latest available for your Linux distribution. For RHEL 6 based distributions, we recommend at least version 1.0.1e-16.el6_5.7.

Use of the Distributed module without the Hadoop module requires that OpenSSH (http://www.openssh.org) be installed and the sshd service be running on all computers. Please install and configure OpenSSH as required for your Linux distribution.

Hadoop Distributions

A prerequisite to running Skytree Server with Hadoop 2 is the installation of one the supported Hadoop distributions on the cluster. Currently supported Hadoop distributions include:

- **MapR**:
  - 3.0.x and 3.1.x
  - 4.0.2 with YARN
  - MapR-FS must be mounted via NFS, and the edition used (M5 or M7) must support this.
- **Cloudera CDH**:
  - 4.4.0, 4.5.0, 4.6.0, 4.7.0, 5.0.x with MapReduce
  - 5.3.x or 5.4.x with YARN
- **Hortonworks HDP**:
  - 1.3.x, 2.0.x, 2.1.x with MapReduce
  - 2.2.x with YARN

Please consult the *Installation for YARN/Hadoop* (page 13) section for further information.

Directory Dependencies

Skytree Server will write a small amount of data to the system's temporary directory. The location of the directory is /tmp by default but will be taken from the TMPDIR environment variable, if set. That directory must be available, and Skytree Server users must have read and write permissions on that directory.

Chapter 3 Installation

This section describes the installation process. Refer to the following topics:

Standalone Installation

Installation on a given host may be performed using either a self-extracting shell script or RPM Package Manager (RPM). Both processes are described in the sections that follow.

Self-Extracting Shell Script Installation

The self-extracting shell script installation can be performed using one of the following method:

- Interactive Installation
- Batch Installation

These are described in the sections that follow.

Interactive Installation

To install Skytree Server interactively on a single machine using the self- extracting shell script, transfer the installer package to that host. We assume it is named `SkytreeServer-<version>.sh`, and we assume that the installation is performed by the system superuser (root or administrative user). See the information below regarding installation as a nonadministrative user.

To verify the integrity of the package without installing it, run

```
# sh SkytreeServer-<version>.sh --check
```

To begin the interactive installation, change to the directory containing the package and, as superuser, type

```
# sh SkytreeServer-<version>.sh
```

The installer first verifies the integrity of the package and extracts it to a temporary directory (in `/tmp`). If successful, the interactive installation begins with a number of prompts given to the user. All prompts have default values given in square brackets (`[]`). The installer prompts for:

The installation path
: The path to Skytree Server's files.

Whether to create a symbolic link to the installation directory
: Answering `yes` creates a symbolic link to the installation directory. This eases the maintenance of multiple simultaneous Skytree Server installations (e.g., to retain old versions) by allowing unique naming for each version's installation directory and a symbolic link that points to the desired default version.

Whether to create symbolic links to the executables
: Answering `yes` creates symbolic links to the installed executables. Creating these in a generally accessible location (e.g., `/usr/bin`) allows users to set their system paths to include that location and easily execute Skytree.

The path for executable links (if necessary)
: The directory where the links described above should be created. The directory must already exist.

The user name for the installation
: The name of the user owning the installation.

The group name for the installation
: The name of the group owning the installation.

Skytree Server can also be installed without superuser (root) privileges. This may be useful when installations are not intended for general availability or the installer does not have administrative privileges for the machine. To install as a regular user, use the `--no-privileged-user (-X)` after the special – separator:

```
# sh SkytreeServer-<version>.sh -- -X
```

If the installer is run as superuser, a `skytree-server` file will be copied to `/etc/bash_completion.d`. This will enable tab-completion for `skytree-server` and `skytree-yarn` commands in zsh and bash.

If the `/tmp` directory is not available or insufficient storage is available for extraction, a different path may be specified by running

```
# sh SkytreeServer-<version>.sh --target <dir>
```

where `<dir>` is the extraction directory.

Batch Installation

When installing on multiple hosts or via automated script, one can bypass the interactive installer described in the previous section. The batch installer can be invoked by specifying the `--no-interactive (-B)` option. This option must be provided *after* the special separator token `--`. Note that all installation parameters can also be specified at the command line after the separator. The available options along with their short option alternatives (in()) and defaults (in []) are:

`--install-dir <directory> (-d) [/opt/skytree/SkytreeServer-<version>]`
: Installation path. Directory will be created if it does not exist.

`--exec-link-dir <directory> (-e) [/usr/bin]`
: Directory in which to create links to binary executables. Directory must already exist.

`--user <user> (-u) [root]`
: User owning the installation. Used only if run as root, ignored otherwise.

`--group <group> (-g) [root]`
: Group owning the installation. Used only if run as root, ignored otherwise.

`--no-exec-link (-L) [false]`
: Do not create symbolic links for the executable files. The default is to create links.

`--no-dir-link (-D) [false]`
: Do not create a symbolic link pointing to the installation directory. Overwrites existing link. The default is to create the link.

`--no-interactive (-B) [false]`
: Run in batch mode, not interactive mode. Use with caution, conflicting files/directories will be overwritten automatically. The default is to run interactively.

`--no-privileged-user (-X) [false]`
: Do not require to be run as root. The default is to run as superuser.

`--help (-h)`
: Display the available options and their defaults, if applicable.

The following command illustrates a batch installation of Skytree Server in the `/path/to/skytree/server/SkytreeServer-latest` directory as group `skytree`. (The remaining options are taken from the default settings.)

```
# sh SkytreeServer-<version>.sh       \
  --                                  \
  --no-interactive                    \
  --install-dir /path/to/skytree/server/SkytreeServer-latest \
  -g skytree
```

Warning: When using the batch installer and running as a privileged (root) user, take care to specify link and directory paths correctly. Incorrect arguments can lead to overwriting existing data on the file system.

RPM Installation

An RPM Package Manager (RPM) package is provided for installation on Red Hat Enterprise Linux 6 and derivative Linux distributions like CentOS and Scientific Linux.

To install the RPM, which we assume is named `SkytreeServer-<version>.rpm` and resides on the installation machine, run the following command as superuser (root):

```
# rpm -ivf SkytreeServer-<version>.rpm
```

This will install Skytree Server to the default location, `/opt/skytree/SkytreeServer-<version>` and place symbolic links to various executables in `/usr/bin`.

For further information on `rpm` usage, try `man rpm` in a terminal window or consult RPM.org (http://www.rpm.org).

Note: The RPM is "relocatable", allowing the `/opt` and `/usr` prefixes to be changed if desired. For example, to install somewhere other than `/opt`, run:
`rpm --relocate /opt=<new path> -ivf SkytreeServer-<version>.rpm`

User Environment Configuration

If, during installation, symbolic links to the installed executables are created and placed in a directory that is not in the user's path, the PATH environment variable can be modified. For BASH users, add the following line to `/home/<username>/.bashrc`:

```
# export PATH=<path to symbolic links>:$PATH
```

For (T)CSH users, add the following line to `/home/<username>/.cshrc`:

```
# setenv PATH <path to symbolic links>:$PATH
```

Note: If the installation was performed without superuser privileges, the `zsh/bash` completion file cannot be copied to `/etc/bash_completion.d`. In that case, a user may source the file available in the installation directory. For BASH users, add the following lines to `/home/<username>/.bashrc`:

```
source <installation path>/config/skytree-server.bash_completion
```

Distributed Installation

Skytree Server's Distributed module enables computation across multiple computers connected via a network (a compute cluster). We describe Skytree Server's installation and configuration for use with the Distributed module here.

There are two important differences between Skytree Server's standalone and distributed installation and configuration that will be described below:

- the requirement of a *Shared File System* (page 10); and
- the need for *SSH via Public/Private Key Authentication* (page 11) (*only for non-Hadoop installations*).

Skytree Server must also be installed in the same manner on all hosts of the cluster, i.e., with all installation paths being the same on each machine. This is most easily accomplished by installing on the *SSH via Public/Private Key Authentication* (page 11) itself but that is not necessary. To install Skytree Server on a shared file system or locally on each host, follow the instructions in *Standalone Installation* (page 7). If installing locally on each host *Batch Installation* (page 8) or *RPM Installation* (page 9) provides an easily repeatable process.

In addition, after installation, a `hosts` file can be found in `<installation path>/config/hosts`. The hostnames or IP addresses of all hosts in the cluster (one per line) may be added there. Skytree Server's `status` module will, by default, report status for all hosts given in that list. Note that execution of Skytree Server jobs will not use the `hosts` file.

Shared File System

Skytree Server's Distributed module requires that a shared file system be available to all Skytree Server users to read and write input and output data files for Skytree Server. In addition, the shared file system must be mounted on all hosts using the same path such that a reference to, e.g., `/path/to/shared/filesystem/file` allows access to the same file from all hosts in the cluster.

SSH via Public/Private Key Authentication

Skytree Server's Distributed module requires that the user be able to `ssh` from any host in the cluster to any other without typing a password or passphrase. We briefly describe the configuration of public/private key authentication for OpenSSH and suggest that you consult its documentation for further information. Logged in as the Skytree user we begin by generating an RSA key pair with `ssh-keygen`. (Note that RSA is preferred over DSA for security reasons.)

```
# ssh-keygen -t rsa
```

Follow the on-screen prompts, leaving the location for the key as the default (`$HOME/.ssh/id_rsa`) and leaving the passphrase empty by typing `Enter` when prompted.

> **Note:** The YARN/Hadoop module removes the dependency on passwordless `ssh` logins.

> **Note:** While we use the more secure RSA keys, we describe only the simplest key exchange, which uses an empty passphrase. More secure public/private key authentication is possible by using an RSA passphrase and the `ssh-agent` program. Please consult the OpenSSH documentation for further information. Note that the `ssh-agent` is required so that no passphrase needs to be typed when `ssh`-ing from one machine to another.

A `$HOME/.ssh/authorized_keys` file containing a public key and a `$HOME/.ssh/id_rsa` private key file must be available to the user on each machine in the cluster. If the home directory resides on a shared file system, you can append the public key to the local `authorized_keys` file:

```
# cat $HOME/.ssh/id_rsa.pub >> $HOME/.ssh/authorized_keys
```

If the machines do not have a shared home directory, you will need to copy the private and public keys to each other machine. You can use secure copy, `scp`, to copy the private and public keys to the other machines. For convenience, OpenSSH distributions often include a script, `ssh-copy-id`, that will append the public key from one machine to the `authorized-keys` file on another.

Once you have installed the private and public keys on the cluster, no password or passphrase should be needed to use `ssh` to connect to any of the configured machines.

> **Note:** When errors occur, they are often due to file permission problems. Ensure that $HOME and $HOME/.ssh have (at least) mode 700 (read, write and execute for the user). Also, the files within the directory should have the following permissions:
>
> ```
> -rw------- authorized_keys
> -rw------- id_rsa
> -rw-r--r-- id_rsa.pub
> ```
>
> To set these permissions, run:
>
> ```
> chmod 644 $HOME/.ssh/id_rsa.pub
> chmod 600 $HOME/.ssh/id_rsa $HOME/.ssh/authorized_keys
> ```

Additional Configuration

Additional configuration settings for Skytree Server can be specified in the (optional) file `<installation_page>/config/skytree.conf`. The available options are explained in the sections that follow.

Thread Configuration

For methods supporting the `--threads` option, global control over the default and maximum number of threads can be specified in the `[threads]` section as follows:

```
[threads]
default_threads=4
max_threads=16
```

With the above example configuration, `--threads=4` is implied, unless otherwise specified by the user. Jobs asking for more than 16 threads will run with 16 threads, and a warning message will be displayed.

Port Configuration

By default, Skytree Server attempts to bind to randomly selected ports numbered between 1024 and 65535 for communication between processes. This behavior can be configured in the `[distributed]` section of the configuration file. The lowest port used is given by `min_port`, and the number of ports used is given by `num_ports`. For example:

```
[distributed]
min_port=1024
num_ports=65411
```

Note: We recommend that `num_ports` be at least 100.

Monitoring Usage

Skytree Server allows you to specify a root directory in `skytree.conf` for storing audit files. For example:

```
[monitor]
monitor_path=/tmp/test-monitor
```

In the above case, Skytree Server audit files will be created in subdirectories of the `/tmp/test-monitor` folder, with each subdirectory named after the current day. For example:

```
admin@node2: /tmp/test-monitor$ ls -1
total 8
drwxrwxr-x 2 mjones mjones 4096 Oct 30 17:08 20151030
drwxrwxr-x 2 mjones mjones 4096 Nov  2 11:22 20151102
```

Within each subdirectory, every Skytree Server run triggers the creation of a uniquely named json file that uses the current date and time and that is appended by the decimal microseconds of the initial timestamp. For example:

```
admin@node2: /tmp/test-monitor/20151102$ ls -l
total 20
-rw-rw-r-- 1 mjones mjones 365 Nov  2 11:06 21051102110623-5691221
-rw-rw-r-- 1 mjones mjones 366 Nov  2 11:14 20151102111437-5300432
-rw-rw-r-- 1 mjones mjones 276 Nov  2 11:17 20151102111736-2082133
-rw-rw-r-- 1 mjones mjones 276 Nov  2 11:20 20151102112041-5001052
-rw-rw-r-- 1 mjones mjones 276 Nov  2 11:22 20151102112214-5142441
```

Each of the above files is written twice: once at the start of the program and once at the (successful) finish. For example, typical content of the file after a successful completion is as follows:

```
admin@node2:/tmp/test-monitor/110215$ cat 110215111437-5300
{
  "user"              : "mjones"
  "hosts"             : "localhost"
  "processesPerHost" : "3"
  "threads"           : "16"
  "workingDirectory" : "/home/mjones/skytree-core-2/server/release.build"
  "command"           : "/home/mjones/skytree-core-
2/server/release.build/bin/skytree-server"
  "startTime"         : "Mon Nov  2 11:14:37 2015"
  "endTime"           : "Mon Nov  2 11:14:47 2015"
  "duration"          : "10"
  "normalExit"        : "true"
}
```

Note: If Skytree Server does not finish successfully, then the contents of the file will include the line: `"abnormalTermination":"true"`

Installation for YARN/Hadoop

This section outlines the installation steps needed to allow execution of distributed Skytree Server jobs on a Hadoop cluster for large-scale enterprise-grade machine learning in a multi-user environment.

There are many advantages of running Skytree Server on a Hadoop cluster, including the following:

- Application of Skytree Machine Learning algorithms on data residing on a Hadoop 2.2 (YARN) or Hadoop 2.0 cluster.
- Job/Application scheduling
- Common programming model with Hadoop MapReduce (Hive, Pig, HBase, etc.)
- Common administration model
- Proximity of data to Skytree Server (replicated on compute nodes)
- High-Availability file system, NameNode and JobTracker (on some distributions)

Execution Model for Hadoop MadReduce v1

Skytree Server runs as a regular Hadoop MapReduce v1 job where the mappers are trivial, and the reducers are complex and execute Skytree Server child processes. Skytree Server does not follow the MapReduce programming paradigm (no map/shuffle/reduce steps), but it still runs as a MapReduce job: It gets scheduled by the JobTracker, and then TaskTrackers launch reducer (Skytree Server) processes that use data nodes as compute nodes to perform machine learning. All of this is done automatically by the Skytree Server Hadoop module, with only minimal changes to the command-line options, while adding quorum (and possibly memory usage) specifications.

Installation Requirements

For Skytree Server to run on a Hadoop cluster, a few basic requirements must be met.

YARN Requirements

The following additional requirements must be met in order to run Skytree Server on YARN.

- **YARN GA version**: YARN went GA with version Hadoop 2.2.0. This is the minimum Hadoop version supported by the Skytree Server YARN application
- **Zookeeper**: Skytree Server requires zookeeper to be installed and accessible to each node in the YARN cluster. Set the following property in the `skytree` configuration file to launch with the Zookeeper connection string:

  ```
  skytree.zookeeper.connection=localhost:2181
  ```
- **HADOOP_CONF_DIR**: This environment variable is generally set during installation. Still, YARN client/edge nodes might not have it set. This can be added to user's bash profile or exported right before launching the YARN application:

  ```
  export HADOOP_CONF_DIR=<Path to yarn installation>/conf
  ```
- **YARN Property**: `yarn.scheduler.maximum-allocation-mb` is used to cap the maximum amount of memory that can be allocated to each container in YARN. Because Skytree Server can be memory intensive, we suggest keeping this value high.

Hadoop 2.0 Requirements

- Identification of a set of cluster nodes to use as compute nodes for Skytree Server. See *Node Allocation* (page 14).
- Skytree Server installed on all compute nodes (as in the distributed case), preferably on a local file system. See *Distributed Installation* (page 10).
- A working MapReduce framework on all compute nodes. See *Installing YARN/Hadoop* (page 15).

Additionally, for MapR clusters only:

- NFS Gateway service running on all compute nodes (those running `TaskTracker`)
- User-writeable MapR volume NFS-mounted on all compute nodes. See *Mount MapR-FS via NFS* (page 16).

The following configuration options are suggested for the compute nodes, see *MapReduce Configuration* (page 16):

- Use of the Fair Scheduler or Capacity Scheduler (recommended)
- Configuring the number of slots for mappers and reducers
- Configuring the memory limit for TaskTrackers The requirements and recommended options are explained in more detail in the following sections.

> **Note:** We suggest that you read this entire section before installing Skytree Server on a Hadoop cluster.

Node Allocation

Due to limitations of the Hadoop scheduling subsystem, you may choose to operate Skytree Server on a dedicated set of cluster nodes. When choosing compute nodes, you should consider the fact that Skytree Server jobs are often more compute- and memory-intense than typical Hadoop MapReduce jobs (where the performance is often limited by disk I/O speed). Also consider using co-located nodes for higher interconnect speeds.

Note: Hardware specification affects overall system performance. High-performance compute, storage and interconnect subsystems can lead to significantly higher throughput of Skytree Server jobs at low additional TCO.

Note: Deployment of Skytree Server jobs on a Hadoop cluster that is simultaneously used for non-Skytree MapReduce jobs may require the use of the Fair Scheduler or the Capacity Scheduler, both of which may result in load sharing restrictions. With the Fair Scheduler on a MapR cluster, it is possible to place jobs on specific nodes or groups of nodes using label-based scheduling. For help, consult http://www.mapr.com/doc/display/MapR/Placing+Jobs+on+Specified+Nodes.

Warning: Inhomogeneous node configurations are strongly discouraged, as they can lead to reduced performance or job failures due to memory exhaustion, load imbalance, and other problems that can be difficult to discover.

Installing YARN/Hadoop

YARN/Hadoop installations can be performed on a variety of platforms. Refer to the following sections to determine the supported platform and for specific recommendations.

- Cloudera's Distribution Including Apache Hadoop (CDH) (page 15)
- Hortonworks Data Platform (HDP) (page 15)
- MapR Editions (page 16)
- Mount MapR-FS via NFS (page 16)
- Kerberos-Enabled Clusters (page 16)
- Running YARN Acceptance Tests (page 22)

Cloudera's Distribution Including Apache Hadoop (CDH)

We recommend installation of CDH4.4 or later using the Cloudera Manager:

http://www.cloudera.com/content/cloudera/en/documentation/cloudera-manager/v4-latest.html

or

http://www.cloudera.com/content/cloudera/en/documentation/cloudera-manager/v5-latest.html

Hortonworks Data Platform (HDP)

For HDP1.3, we recommend installation using Apache Ambari (http://docs.hortonworks.com/HDPDocuments/HDP1/HDP-1.3.2/bk_using_Ambari_book/content/index.html).

For HDP2.x, we recommend installation using Apache Ambari (http://docs.hortonworks.com/HDPDocuments/HDP2/HDP-2.0.5.0/bk_using_Ambari_book/content/index.html).

MapR Editions

We recommend following the MapR Installation Guide (http://mapr.com/doc/display/MapR/Installation+Guide) to install a MapR edition. Skytree requires only the MapReduce framework and the NFS API to be operational, therefore any MapR edition (M3, M5 or M7) may be used.

Depending on the cluster size, a small number of nodes might be reserved to run only services other than `TaskTracker`. For small clusters, it is often possible and reasonable that all nodes serve as data and compute nodes. Below is a list of services that are required:

- The following MapR services must run on all Skytree Server compute nodes:
  - `FileServer`
  - `NFS Gateway` for Skytree Server to read/write data, see *Mount MapR-FS via NFS* (page 16)
  - `CLDB`
  - `TaskTracker` for MapReduce framework to launch Skytree Server, see *MapReduce Configuration* (page 16)
  - `HostStats`
- The `JobTracker` service must run on at least one cluster node (for job submission)
- The `WebServer` service must run on at least one cluster node (for administration)

Mount MapR-FS via NFS

MapR installation configures data nodes such that their local disk storage is part of the MapR-FS, see http://www.mapr.com/doc/display/MapR/Disks. This has the effect of making these disks available for MapReduce jobs as shared distributed storage via HDFS (in hdfs://user/<username>/). Skytree Server, however, requires this same shared distributed storage to be made available via the `NFS Gateway` service. Accordingly, the `NFS Gateway` service must be configured to run on all compute nodes, and each node's `localhost:/mapr` should be mounted to `/mapr` via the NFS protocol. Then, the user's Skytree Server jobs will be able to access shared data stored in, e.g., the user's home directory at `/mapr/<cluster name>/user/<username>/`. For help, consult http://www.mapr.com/doc/display/MapR/Setting+Up+the+Client#SettingUptheClient-nfs.

The replication factor of the MapR volumes may be adjusted for performance or HA considerations.

Kerberos-Enabled Clusters

Users will need to generate forwardable Kerberos ticket-granting tickets (TGTs). These can be generated using `kinit -f`. To make this the default behavior of `kinit`, set `forwardable = true` in the `[libdefaults]` section of `/etc/krb5.conf`.

Tickets will be read when the job is submitted. Because some processes can be time consuming, the ticket must be renewable throughout the duration of job execution.

MapReduce Configuration

Skytree Server runs as a MapReduce job with trivial mapper tasks and complex reducer tasks that each spawn a Skytree Server process. We recommend the Hadoop schedulers are the Fair Scheduler (http://hadoop.apache.org/docs/stable/fair_scheduler.html) and the Capacity Scheduler (http://hadoop.apache.org/docs/stable/capacity_scheduler.html).

Resource sharing between users (or groups) is achieved with either the Fair or the Capacity Scheduler, where the number of Skytree Server processes (and hence the number of reducer slots) per compute node is controllable.

> **Note:** Use of the Capacity Scheduler is required when running under Hadoop 2.0 with YARN.

> **Warning:** In contrast to regular MapReduce jobs, Skytree Server jobs may require a majority of the node's available memory. Thus, it is not advisable to have multiple Skytree Server processes competing for memory on the same compute node unless the Capacity Scheduler is used, and the reserved amount of memory per Skytree Server process is specified by the user.

> **Note:** If a FIFO scheduler (such as the Capacity Scheduler) is used, it may be possible to submit multiple jobs at once, which can delay the execution of other user's jobs submitted shortly thereafter.

> **Note:** In addition to having a properly configured scheduler in place, Skytree Server jobs should be submitted with a quorum condition that allows operation on a limited set of nodes for increased throughput of jobs in shared environments.

Fair Scheduler

If the Fair Scheduler is employed, each Skytree Server process will occupy one reducer slot. For best performance it is strongly recommended to limit the number of Skytree Server processes (and hence the number of reducer slots) per compute node, and to allow Skytree Server to use all the available memory on the compute node. Under the Fair Scheduler, the only way to guarantee exclusive access to a node is to set the number of slots to 1.

To schedule Skytree Server jobs with the Fair Scheduler, we suggest you modify the MapReduce configuration file `mapred-site.xml` on each node (running `JobTracker` or `TaskTracker`) as follows:

```
<property>
   <name>mapred.jobtracker.taskScheduler</name>
   <value>org.apache.hadoop.mapred.FairScheduler</value>
</property>
<property>
   <name>mapred.tasktracker.map.tasks.maximum</name>
   <value>1</value>
</property>
<property>
   <name>mapred.tasktracker.reduce.tasks.maximum</name>
   <value>1</value>
</property>
<property>
   <name>mapreduce.tasktracker.reserved.physicalmemory.mb</name>
   <value>48000</value>
</property>
```

This configures one mapper and one reducer slot, and sets the maximum amount of memory that Skytree Server (a memory-hungry child of `TaskTracker`) is allowed to use on each node. Replace 48000 with a value that corresponds to 75% of system memory (in MB) per node. This allows other processes to co-exist while Skytree Server jobs are running (e.g., MapR-FS on MapR, system processes, etc.).

Warning: Using the Fair Scheduler with more than one reducer slot per node can lead to resource conflicts (especially memory starvation) between multiple Skytree Server processes. We discourage this mode of operation. However, if absolutely needed, it might make sense to also globally limit the number of threads per process to improve resource sharing, refer to *Thread Configuration* (page 12).

Note: On MapR systems, the MapR-FS service uses 20% of the system memory for file system management by default. Hence, we use a default value of 75% of system memory as the maximum amount of memory to be used by each TaskTracker (and its children, i.e., Skytree Server). Both values can be adjusted for performance optimization.

Note: Modifying `mapred-site.xml` requires a restart of the `JobTracker` and/or `TaskTracker` services.

Capacity Scheduler

If the Capacity Scheduler is employed, each compute node may be configured to have multiple mapper and reducer slots, and each slot may be associated with a partial amount of the node's available memory. Multiple Skytree Server jobs (and/or processes) can then run simultaneously on a node by sharing its virtual memory. Each Skytree Server process occupies as many reducer slots as needed for the reserved amount of memory specified by the user.

To schedule Skytree Server jobs with the Capacity Scheduler, we suggest you modify the MapReduce configuration file `mapred-site.xml` on each node (running `JobTracker` or `TaskTracker`) as follows:

```
<property>
   <name>mapred.jobtracker.taskScheduler</name>
   <value>org.apache.hadoop.mapred.CapacityTaskScheduler</value>
</property>
<property>
   <name>mapred.cluster.map.memory.mb</name>
   <value>2000</value>
</property>
<property>
   <name>mapred.cluster.reduce.memory.mb</name>
   <value>2000</value>
</property>
<property>
   <name>mapred.cluster.max.map.memory.mb</name>
   <value>48000</value>
</property>
<property>
   <name>mapred.cluster.max.reduce.memory.mb</name>
   <value>48000</value>
</property>
<property>
   <name>mapred.tasktracker.map.tasks.maximum</name>
   <value>24</value>
</property>
<property>
   <name>mapred.tasktracker.reduce.tasks.maximum</name>
   <value>24</value>
</property>
<property>
```

```
    <name>mapreduce.tasktracker.reserved.physicalmemory.mb</name>
    <value>48000</value>
</property>
```

As in the Fair Scheduler configuration above, we assume that there are 48000 MB of available memory for the `TaskTracker` (75% of system memory) per node. Instead of 1 slot (of size 48000 MB), the Capacity Scheduler allows us to use 24 slots of 2000 MB each for improved resource sharing granularity. Replace 48000 with a value that corresponds to 75% of system memory (in MB) per node, and replace 24 with the desired number of slots. Adjust the resulting memory per slot (2000) accordingly.

Note: The sample configuration above allows Skytree Server processes to exclusively occupy a compute node by reserving all available memory (and hence all slots). Specify lower values for `mapred.cluster.max.{map,reduce}.memory.mb` to limit memory usage of Skytree Server. In addition to reducing memory usage, it may also be helpful to limit the number of threads per process. Refer to *Thread Configuration* (page 12).

Note: The Capacity Scheduler allows configuration of multiple job queues and specification of job priorities. This can be used to improve resource sharing. Refer to the documentation of the Capacity Scheduler (http://hadoop.apache.org/docs/stable/capacity_scheduler.html).

Note: By default, the MapR-FS service uses 20% of the system memory for file system management. Hence, we use a default value of 75% of system memory as the maximum amount of memory to be used by each TaskTracker (and its children, i.e., Skytree Server). Both values can be adjusted for performance optimization.

Note: Modifying `mapred-site.xml` requires restart of the `JobTracker` and/or `TaskTracker` services.

Hadoop Classpath Configuration

For MapR and Cloudera, we suggest you add the following line to `hadoop-env.sh` on each node:

```
export HADOOP_CLASSPATH=\
  <installation path>/lib/skytree-server-hadoop.jar:$HADOOP_CLASSPATH
```

For Hortonworks BASH users, add the following line to `/home/<username>/.bashrc`:

```
export HADOOP_CLASSPATH=\
  <installation path>/lib/skytree-server-hadoop.jar:$HADOOP_CLASSPATH
```

For Hortonworks TCSH users, add the following line to `/home/<username>/.cshrc`:

```
setenv HADOOP_CLASSPATH \
  <installation path>/lib/skytree-server-hadoop.jar:$HADOOP_CLASSPATH
```

Warning: Omitting the HADOOP_CLASSPATH can lead to Skytree Server runtime errors, such as 'Error: Could not find or load main class com.skytree.SkytreeServer'.

Note: Modifying `hadoop-env.sh` requires restart of the `JobTracker` and `TaskTracker` services.

Hadoop Properties File

Hadoop configuration settings for Skytree Server can be specified in the (optional) `<installation path>/config/skytree-hadoop.properties` file. Available options are explained in the sections that follow.

Zookeeper Settings

For Apache Hadoop setup, Skytree Server uses Zookeeper for cluster coordination. This property is required in that scenario.

```
skytree.zookeeper.connection=localhost:2181
```

Quorum Settings

Default cluster-wide quorum settings, used if a user omits the `-Dquorum` option, can be specified using the `skytree.job.quorum` option, e.g.:

```
skytree.job.quorum=5,10m;3,1h;1,4h
```

A maximum quorum setting may also be configured via `skytree.job.quorum`.max to limit the amount of time users' jobs may wait for compute resources. For example, the setting below allows users to specify quorums that wait up to 10 minute for 10 processes, up to 45 minutes for 5 to 9 processes and up to 4 hours for 4 or fewer processes:

```
skytree.job.quorum.max=10,10m;5,45m;1,4h
```

Default HDFS Connection Settings

If the data to be accessed is on a remote HDFS server, its hostname and port can be specified as the default. If nothing is specified, then hostname will be `default` and port 0.

```
skytree.hdfs.host.name=localhost
skytree.hdfs.port=8020
```

Hadoop Library Paths

Explicit paths to `libhdfs.so` and `libjvm.so` should be provided only if Skytree fails to automatically locate the libraries.

```
skytree.hadoop.libhdfs=/usr/lib64/libhdfs.so
skytree.hadoop.libjvm=/usr/lib64/libjvm.so
```

Note: For Cloudera installations via the Cloudera Manager, `libhdfs.so` may be located in a parcel directory (e.g., `/opt/cloudera/parcels/CDH/lib64/libhdfs.so`). The locate utility may be helpful in determining the path to the library:

```
locate libhdfs.so
```

YARN Properties File

YARN configuration settings for Skytree Server can be specified in the (optional) `<installation path>/config/skytree-yarn.properties` file. Suggested properties are explained in the sections that follow.

MapR 4 Settings

In order for `skytree-yarn` to work with MapR 4, the following properties should be updated.

- Add the path to `mapr.login.conf`. This is usually in `/opt/mapr/conf`.

  `skytree.jass.config.path=<path/to/mapr.login.conf>`

- Add the MapR NFS mount path. This is usually in `/mapr/<cluster_name>`.

  `skytree.mapr.nfs.mount.path=</maprfs/nfs/mount/path>`

Monitor Settings

Admins can specify a directory for `skytree.monitor.dir` property.

```
skytree.monitor.dir=<directory_path>
```

If this is set, then a subdirectory is created each day using the naming convention "MMDDYY", and a single file per executing of `skytree-yarn` is written to that subdirectory.

As an example:

```
$ tree /var/log/skytree
/var/log/skytree
-- 032015
    -- skytree-032015231713-4Z8TCM5T
    -- skytree-032015231749-91JIHZ5L
    -- skytree-032015235128-CBL166CJ
    -- skytree-032015235204-4P6IWYZ5

1 directory, 4 files
$ cat /var/log/skytree/032015/skytree-032015235204-4P6IWYZ5
{
    "user": "skytree",
    "host": "myhost",
    "arguments": "gbt --training_in /user/skytree/datasets/income.data.st \
                 --training_labels_in /user/skytree/datasets/income.data.labels \
                 --num_trees 10 \
                 --model_out model \
                 --mem 4000 \
                 --processes 1 \
                 --cores 2 \
                 --output /user/skytree/core-1376-test-2 \
                 --loglevel debug",
    "resourceRequest": {
    "processes": 1,
    "cores": 2,
    "memory": 4000
```

```
    },
    "submitTime": "Fri Mar 20 23:52:04 UTC 2015",
    "endTime": "Fri Mar 20 23:52:39 UTC 2015"
}
```

Note: If the `skytree-yarn` execution is aborted, the `endTime` will be empty. Aborting `skytree-yarn` does not necessarily mean that the YARN job was killed.

Running YARN Acceptance Tests

Users running Skytree Server on YARN can optionally run a script that performs a series of YARN acceptance tests. This script is available in the `SkytreeServer/bin` directory and runs the following tests:

1. Verifies the existence of the Zookeeper zNode
2. Checks basic HDFS and YARN functionality
3. Runs the `skytree-server status` command
4. Runs a basic machine learning problem using Skytree Server

The following command runs the YARN acceptance tests:

```
yarn_validation.sh <options>
```

The following table describes the options that are available with this script:

Table 1: *YARN Validation Options*

Command	Description
`-c --cores`	Specify the number of threads to use per process when testing Skytree Server. This value defaults to 2.
`-d --test_dir`	Specify the directory in HDFS to use as scratch space for testing. This defaults to `./SkytreeServer-Test`.
`-h --help`	Prints the help output.
`-H --hdfs_path`	Specify the path to the HDFS executable (if not automatically detected).
`-m --memory`	Specify the memory allocation in Mb for each YARN container. This value defaults to 2000.
`-p --processes`	Specify the number of YARN containers to use for testing. This value defaults to 4.
`-y --yarn_path`	Specify the path to the YARN executable (if not automatically detected).
`-z --zookeeper_path`	Specify the path to the Zookeeper zkCli.sh (if not automatically detected).

Chapter 4 Verifying the Installation

This section describes how to verify that the installation was successful. The process varies depending on the type of installation you performed. Refer to the following topics:

Standalone Verification

Log in as the unprivileged Skytree Server user, and from any directory run:

```
# skytree-server
```

The command displays a list of all available modules.

Finally, running:

```
# generate-header.sh
```

should list the options available for that application.

> **Note:** If you did not create symbolic links to the executables during the installation, or you chose not to put the location of those links in the user's path, you will need to prefix the commands above with the path (relative or absolute) to the executables. For example, if you've installed to `/path/to/skytree/server`, you would run `/path/to/skytree/server/skytree-server`.

Distributed Verification

To verify that Skytree Server is properly configured to run with the Distributed module, verify the standalone installations as described in the previous section (*Standalone Verification* (page 23)), and then use the `status` module to verify distributed operation.

If the `hosts` file was configured (see *Distributed Installation* (page 10)), run:

```
# skytree-server status
```

If not, run with an appropriate list of comma-separated hosts:

```
# skytree-server status --hosts <host list>
```

The output should report information regarding the status of the hosts and any active Skytree Server processes.

Hadoop Verification

To verify that Skytree Server is properly configured to run with the YARN/Hadoop module, issue the following command (as a regular user):

```
# hadoop com.skytree.SkytreeServer      \
  -Djob_dir      "/<path-to-fs>/job"    \
  -Doptions      "status"               \
  -Dquorum       "<number of nodes>,1m" \
  [-Dmem_per_process=2000]
```

Replace `<path-to-fs>` with the path to the user's home directory on the Hadoop Distributed File System (HDFS).

- For MapR installations, `<path-to-fs>` is typically specified as `/mapr/<clustername>/user/<username>/` (files are accessed via the MapR NFS Gateway).
- For other Hadoop installations (such as Cloudera or Hortonworks), `<path-to-fs>` is typically `/user/<username>/`. (Files are accessed via the Skytree HDFS connector.)

Replace `<number of nodes>` with the number of nodes running the `TaskTracker` service (or a subset thereof).

The `-Dmem_per_process` option is required only if the Capacity Scheduler is configured and otherwise disallowed.

A job directory `job` should be created in `<path-to-fs>`, and it should contain a log file called `skytree-server.log`. The log should report information regarding the status of the nodes and any active Skytree Server processes. The job should also be recorded on the JobTracker's web interface (usually at `http://<JobTracker hostname>:50030`).

Note: The above command assumes that the requested number of nodes will become available within 1 minute. Also, the `job` directory must not exist. 2000 MB of reserved memory should be sufficient, if running under the Capacity Scheduler. Increase if needed.

Chapter 5 Upgrading Skytree Server

This section describes how to upgrade Skytree Server. The process varies depending on whether you perform a shell script upgrade or an rpm upgrade. Refer to the following topics:

Before You Begin

Determine the method that you used to install Skytree Server. The process for upgrading will be the same.

- If you used the Self-Extracting Shell Script installation method, then refer to *Self-Extracting Shell Script Upgrade* (page 25) for instructions on how to upgrade Skytree Server.
- If you used the RPM installation method, then refer to *RPM Upgrade* (page 25) for instructions on how to upgrade Skytree Server.

> **Note:** If you are not certain which installation method was used, then try running the following RPM command:
>
> ```
> # rpm -qa | grep SkytreeServer
> ```
>
> If the Skytree Server version is successfully returned, then you can assume that the RPM installation process was used.

Self-Extracting Shell Script Upgrade

If the previous installation used the self-extracting shell script, the new installation should do so as well. The new installation can be installed alongside the old version by following the instructions in *Self-Extracting Shell Script Installation* (page 7). Note that the Skytree Server symbolic link to the installation package, if used, will be updated to point to the new installation.

RPM Upgrade

In order to upgrade an RPM installation, the existing RPM must first be uninstalled. Refer to *Uninstalling the RPM* (page 28) for instructions on how to remove the existing RPM. The new RPM can then be installed by following the instructions in the *RPM Installation* (page 9) section.

Chapter 6 Uninstalling Skytree Server

This section describes how to uninstall Skytree Server. The process varies depending on whether you uninstall using a shell script or using the rpm. Refer to the following topics:

Before You Begin

Determine the method that you used to install Skytree Server. The process for uninstalling will be the same.

- If you used the Self-Extracting Shell Script installation method, then refer to *Self-Extracting Shell Script Uninstallation* (page 27) for instructions on how to remove Skytree.
- If you used the RPM installation method, then refer to *Uninstalling the RPM* (page 28) for instructions on how to remove Skytree Server.

Note: If you are not certain which installation method was used, then try running the following RPM command:

```
# rpm -qa | grep SkytreeServer
```

If the Skytree Server version is successfully returned, then you can assume that the RPM installation process was used.

Self-Extracting Shell Script Uninstallation

If the self-extracting shell script was used for installation, the package can be removed by deleting the installation directory. For example, if Skytree Server was installed to `/path/to/skytree/server/SkytreeServer-latest`, it can be removed using:

```
rm -rf /path/to/skytree/server/SkytreeServer-latest
```

Note: If any configuration files were modified, remember to back up the files e.g., in `/path/to/skytree/server/SkytreeServer-latest/config` before you uninstall Skytree Server.

Symbolic links to the installation directory can also be deleted. The paths to those links are those given at installation time.

Uninstalling the RPM

Before you can uninstall the Skytree Server RPM, you must first determine the version that is currently installed. This information can be obtained by running the following command:

```
rpm -qa | grep SkytreeServer
```

Once the version is determined, run the following command to uninstall the Skytree Server RPM:

```
rpm -e SkytreeServer-<version>
```

where the installation RPM was named `SkytreeServer-<version>.rpm`.

Note that any modified configuration files in the installation directory will be preserved and, if desired, can be manually removed.